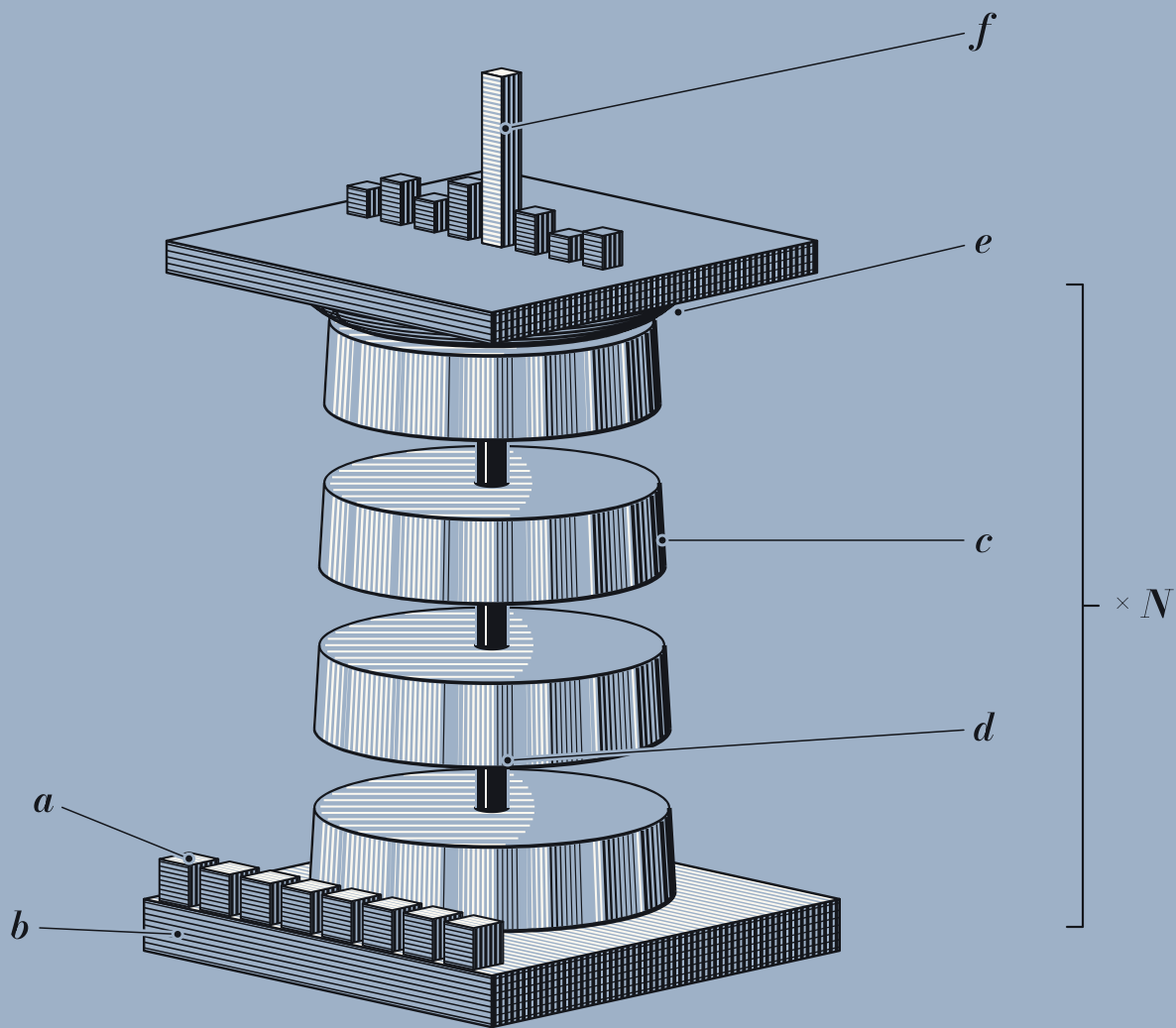


Build Large Language Models (LLMs) from Scratch



*Fig. 1. A decoder-only model, raised as a column. **a** tokens; **b** embedding; **c** a transformer block, one of N ; **d** the residual stream they are threaded on; **e** output head; **f** next-token distribution.*